# Amrita School of Computing

Amritapuri Campus

Department of Computer Science and Engineering

**B.Tech Computer Science and Engineering**

**23CSE498 – Project Phase II**

## Design and Solution Submission

**Project Title** Explainable AI-Based Element Recommendation Framework for Functional Materials Using Unsupervised Representation Learning

**Project Guide** Ms. Remya V S

### Team Members

| Team No. | Roll Number | Name of Student |
|---|---|---|
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Academic Year: 2026–2027

**Amrita Vishwa Vidyapeetham**

# 1 Sustainable Development Goal (SDG) Mapping

**Primary SDG:** SDG 9 — Industry, Innovation and Infrastructure

**Secondary SDG(s) (Optional):** SDG 7 — Affordable and Clean Energy

**Justification (100–150 words):**

The project primarily supports SDG 9 because it is, at its core, a piece of research infrastructure: an AI-driven tool that speeds up materials discovery by replacing slow, expensive trial-and-error experimentation and DFT-only workflows with a similarity-driven, explainable recommendation system. This directly supports the SDG 9 targets of enhancing scientific research and upgrading the technological capabilities of the materials-science sector. It also aligns with SDG 7, since the functional materials the framework targets – batteries, semiconductors, and catalysts – feed directly into energy storage and energy-harvesting technologies. By narrowing down promising substitute elements before expensive experimental or computational validation, the framework is intended to help researchers explore candidate materials for clean-energy applications more efficiently, without claiming to replace that validation step itself.

# 2 Project Milestones and Timeline

| Semester | Major Activity / Milestone | Expected Deliverable | Expected Completion | Status |
|---|---|---|---|---|
| S6 | Project domain exploration | Problem domain identified | 15 Feb 2026 | Completed |
| S6 | Literature review | Initial literature survey | 15 Mar 2026 | Completed |
| S6 | Baseline implementation | Reference implementation completed | 30 Apr 2026 | Completed |
| S7 (Submission 1) | Literature survey | Comprehensive literature survey | 15 Jul 2026 | Completed |
| S7 (Submission 1) | Research gap identification | Research gap statement | 20 Jul 2026 | Completed |
| S7 (Submission 1) | Problem statement | Finalized objectives | 25 Jul 2026 | Completed |
| S7 (Submission 2) | System design | Architecture and workflow | 06 Aug 2026 | Current |
| S7 | Prototype implementation | Functional prototype | 15 Oct 2026 | Planned |
| S7 | Module integration | Integrated system | 15 Nov 2026 | Planned |
| S7 | Testing and validation | Experimental results | 15 Dec 2026 | Planned |
| S8 | System enhancement | Improved system | 31 Jan 2027 | Planned |
| S8 | Performance evaluation | Performance analysis | 28 Feb 2027 | Planned |
| S8 | Research publication | Paper submission | 20 Mar 2027 | Planned |
| S8 | Dissertation writing | Final report | 10 Apr 2027 | Planned |
| S8 | Final demonstration | Presentation and viva | 20 Apr 2027 | Planned |

Table 1: Overall Project Timeline

# 3 System Architecture (If Applicable)

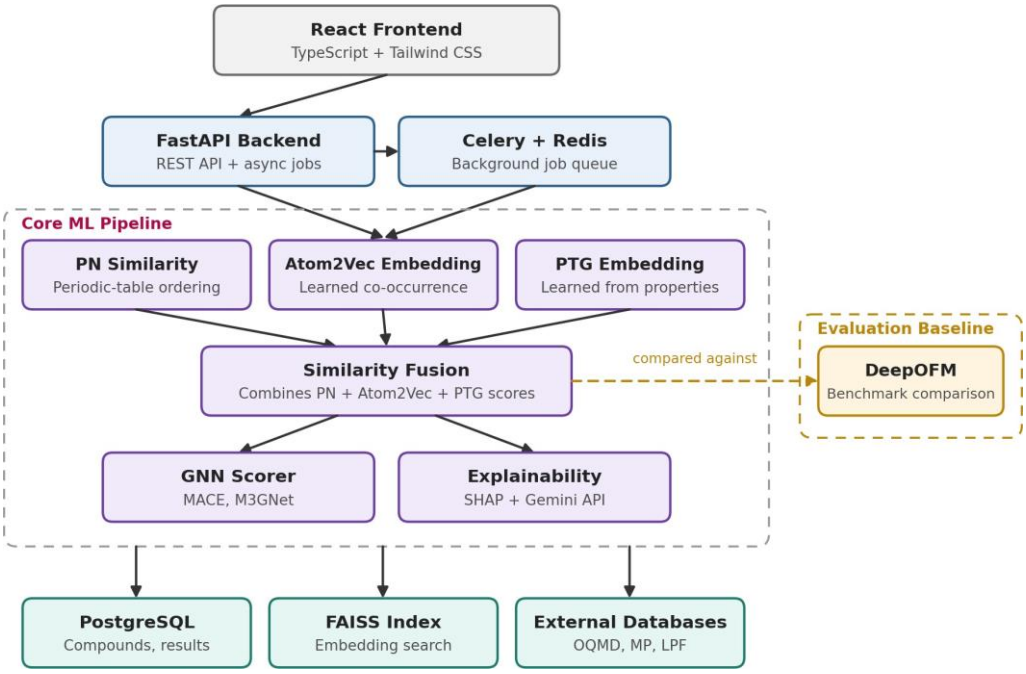



Figure 1: Proposed four-tier system architecture

## Architecture Description

The system follows a four-tier layered architecture. The presentation tier is a React and TypeScript frontend, styled with Tailwind CSS, through which a user submits a target material composition and the specific element to be replaced, and later views the ranked recommendations. The service tier is built on FastAPI, which exposes the REST endpoints and hands off longer-running work – similarity search, GNN scoring, database validation – to a Celery worker pool backed by Redis, so the API does not block while this computation runs.

The core ML pipeline tier is where the scientific logic lives. It consists of three cooperating modules: a similarity engine that combines periodic-number ordering, Atom2Vec-style compositional embeddings, and property-based embeddings to identify chemically comparable elements; a GNN scorer that uses pretrained graph neural network potentials (MACE, M3GNet) to estimate formation energy for candidate substitutions without running full DFT for every candidate; and an explainability module that combines SHAP-based attribution with the Gemini API to turn the numeric scores into a readable justification.

The data tier persists structured results in PostgreSQL, maintains a FAISS index for fast nearest-neighbour lookups over the element and compound embeddings, and pulls reference structures from external materials databases – OQMD, the Materials Project, and the Linus Pauling File – for cross-validation. At this stage of the project, this architecture reflects the design we intend to implement; the modules are not yet fully built out, and the diagram should be read as the target structure rather than a completed system.

# 4 Module Description (If Applicable)

| Module | Description |
|---|---|
| Data Ingestion & Preprocessing | Pulls elemental property data (Mendeleev, XenonPy) and compound/structure data from OQMD, the Materials Project, and the Linus Pauling File, and normalises it into a common schema that the rest of the pipeline can consume. |
| Element Similarity Engine | Computes three independent similarity signals between elements – periodic-number distance, Atom2Vec-style co-occurrence embeddings, and property-based embeddings – and fuses them into a single similarity score used to shortlist candidate substitutes. |
| Candidate Generation & Scoring | Substitutes each shortlisted candidate element into the target compound and scores the resulting structure using pre-trained GNN interatomic potentials (MACE, M3GNet) to estimate formation-energy plausibility. |
| Validation Module | Cross-checks surviving candidates against known compounds in OQMD and the Materials Project using structural similarity tools (pymatgen StructureMatcher, CrystalNN), and flags cases that require further computational or experimental validation. |
| Explainability & Recommendation Interface | Combines the similarity, energy, and database-evidence signals into a confidence score, generates a natural-language justification for each recommendation, and presents the ranked results through the web interface. |

# 5 System Workflow (If Applicable)

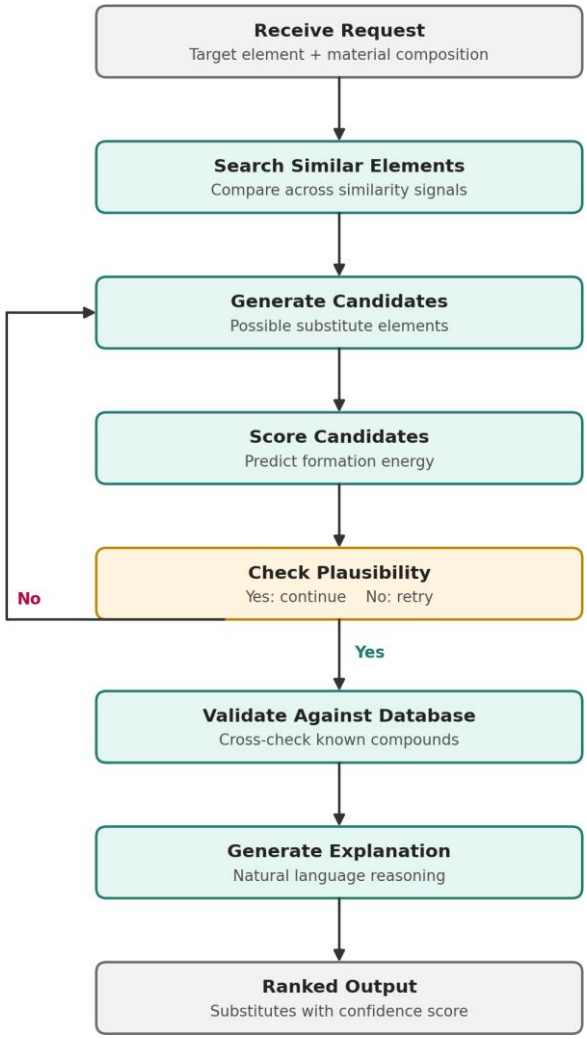



Figure 2: Operational workflow of the recommendation pipeline

## Workflow Description

A request enters the system with a target material composition and the element that needs a substitute. The similarity engine searches for chemically comparable elements and generates a candidate list, and each candidate is then scored for formation-energy plausibility using the GNN scorer. This feeds a plausibility check: if the predicted energy indicates an unstable or implausible compound, the candidate is discarded and the pipeline loops back to generate the next batch of candidates rather than surfacing an unstable substitution – a filtering pattern similar to the thermodynamic-stability checks used in the periodic-number-based crystal structure prediction literature we reviewed.

Candidates that pass are cross-checked against known-compound databases for existing precedent, run through the explanation module, and returned as a ranked list, each with a confidence score and a short natural-language reason for why it was recommended.

# 6 Domain-Specific Design (If Applicable)

| Design Component | Details / Reference |
|---|---|
| **Database Schema / ER Diagram** | Relational schema (PostgreSQL) with tables for elements, compounds, similarity scores, candidate substitutions, and confidence/explanation records. Full ER diagram to be finalised once the schema is implemented. |
| **UML Diagrams** | Class and sequence diagrams for the recommendation pipeline will be developed during the implementation phase, once module interfaces are finalised. |
| **Machine Learning Model Architecture** | No models are trained from scratch for the recommendation pipeline itself. The similarity engine fuses three embedding sources (PN ordering, Atom2Vec-style co-occurrence embeddings, property-based PTG embeddings), and formation-energy scoring uses pretrained GNN interatomic potentials (MACE, M3GNet) as a fast substitute for full DFT calculations. DeepOFM is used only as an independent baseline model for comparative evaluation and is not part of the production feature-extraction or fusion pipeline. |
| **Dataset Description** | Elemental property data from Mendeleev and XenonPy; compound and crystal structure data from the Open Quantum Materials Database (OQMD), the Materials Project, and (where accessible) the Linus Pauling File. |
| **API Design** | FastAPI REST endpoints, e.g. POST /recommend (submit target material and element), GET /candidates/{id} (retrieve ranked candidates), GET /explanation/{id} (retrieve the reasoning for a given recommendation). |
| **Network Topology** | Not applicable – the system is a single web application (frontend, API, workers, database) and does not involve a distributed network topology. |
| **Security Architecture** | Not a major design driver at this stage – the system is an academic research prototype and does not handle personal or sensitive data. API keys (e.g. for the Gemini API) will be managed through environment variables rather than hard-coded. |
| **Hardware Block Diagram** | Not applicable – this is a software-only project; hardware requirements for development and deployment are listed separately in Section 9. |
| **Deployment Architecture** | Backend, database, and worker services are containerised with Docker and orchestrated using Docker Compose for lo- |

# 7 Algorithms / Methodology

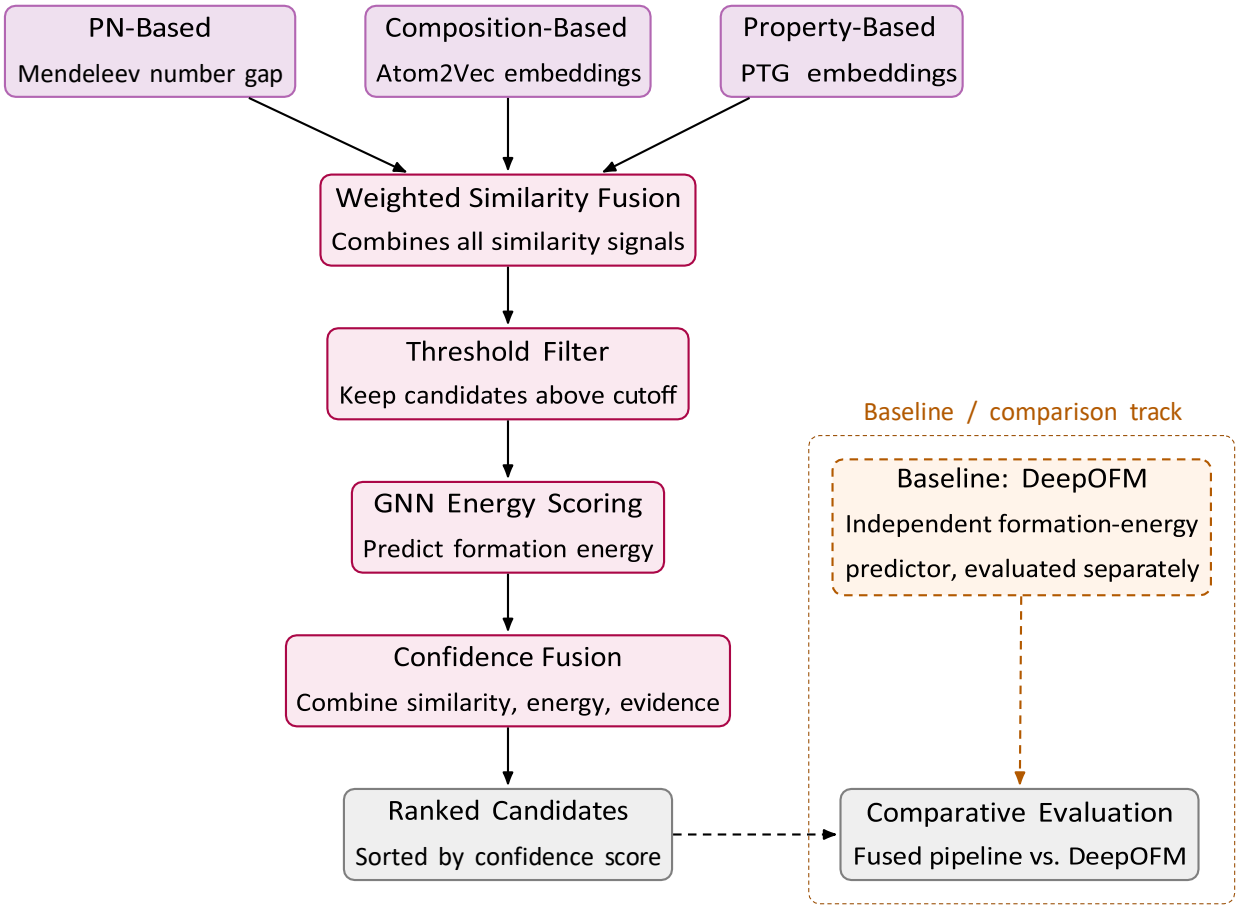



Figure 3: Proposed similarity-fusion and confidence-scoring pipeline. DeepOFM is kept outside the fusion stage and used only as an independent baseline for evaluation.

## Proposed Algorithm / Methodology Description

The core algorithm has three phases.

**Relationship to DeepOFM.** DeepOFM is not treated as a fourth feature-extraction signal inside the similarity-fusion stage. Unlike the PN, Atom2Vec, and property-based (PTG) signals, which are designed to be fused into a single similarity score, DeepOFM is an end-to-end learned predictor. It is kept as an independent baseline, run separately on the same candidate substitutions, and its predictions are compared against the output of our fused pipeline during evaluation (Section 11). This keeps the core pipeline's three similarity signals conceptually homogeneous while still allowing benchmarking against a stronger learned baseline.

**Phase 1 – Multi-signal similarity computation.** For a target element that needs

a substitute, the system computes three independent similarity signals against every other element: a periodic-number (PN) based signal, using the inverse distance between elements on the Mendeleev periodic-number ordering; a composition-based signal, using cosine similarity between Atom2Vec-style embeddings learned from co-occurrence statistics across known compounds; and a property-based signal, using cosine similarity between embeddings derived from raw physicochemical features such as melting point, electronegativity, and atomic radius. These three signals were chosen because they capture different, complementary notions of chemical similarity – periodic-table position, real-world co-occurrence patterns, and continuous physical properties – and combining them is one of the specific gaps we identified in the individual methods reviewed in our literature survey, none of which combine more than one similarity source.

**Phase 2 – Candidate generation and energy filtering.** The three signals are fused into a single similarity score per element pair, and elements above a similarity threshold are shortlisted as candidates. Each candidate is substituted into the target compound, and a pretrained GNN interatomic potential estimates the formation energy of the resulting structure as a fast, DFT-free plausibility check. Candidates with unfavourable predicted energy are filtered out and the candidate pool is regenerated.

**Phase 3 – Confidence scoring and explanation.** Surviving candidates are checked against real compound databases for existing precedent. A final confidence score combines the similarity score, the energy plausibility, and the database evidence into a single value, and this is passed to a language model that generates a short, human-readable justification for the recommendation. Producing this natural-language explanation, rather than stopping at a numeric score, is the main methodological contribution we are aiming for in this project.

## Pseudocode / Mathematical Formulation (Optional)

**Similarity fusion.** For target element $e_t$ and candidate element $e$:

$$S_{PN}(e_t, e) = \frac{1}{1 + |\mathrm{PN}(e_t) - \mathrm{PN}(e)|}$$

$$S_{comp}(e_t, e) = \cos\ \vec{v}_{comp}(e_t), \vec{v}_{comp}(e)$$

$$S_{prop}(e_t, e) = \cos\ \vec{v}_{prop}(e_t), \vec{v}_{prop}(e)$$

$$S(e_t, e) = w_1 S_{PN} + w_2 S_{comp} + w_3 S_{prop}, \qquad w_1 + w_2 + w_3 = 1$$

**Confidence scoring.** For a substituted compound $M'$ (the target compound $M$ with $e_t$ replaced by candidate $e_c$):

$$\text{Conf}(M') = \alpha \cdot S(e_t, e_c) + \beta \cdot \sigma - \Delta E(M') + \gamma \cdot D(M')$$

where $\Delta E(M')$ is the GNN-predicted formation energy, $\sigma$ is a sigmoid mapping favourable (negative) energy to a higher score, and $D(M')$ is a database-evidence term obtained from structural matching against OQMD / Materials Project.

**Pseudocode**

```
function RecommendSubstitutes(material M, target_element e_t, top_k):
    candidates = []
    for e in ElementSet \ {e_t}:
        S_pn   = pn_similarity(e_t, e)
        S_comp = cosine(atom2vec[e_t], atom2vec[e])
        S_prop = cosine(prop_embed[e_t], prop_embed[e])
        S = w1*S_pn + w2*S_comp + w3*S_prop
        if S > tau:
            candidates.append((e, S))

    sort candidates by S descending
    shortlist = candidates[:top_k]

    results = []
    for (e_c, S) in shortlist:
        M_prime = substitute(M, e_t, e_c)
        dE = gnn_formation_energy(M_prime)
        if dE > energy_cutoff:
            continue                          # retry: skip, re-rank next batch
        D = database_match_score(M_prime)     # via OQMD / MP / StructureMatcher
        conf = alpha*S + beta*sigmoid(-dE) + gamma*D
        explanation = generate_explanation(e_t, e_c, S, dE, D)
        results.append({candidate: e_c, confidence: conf,
                        explanation: explanation})

    sort results by confidence descending
    return results
```

The weights ($w_1$, $w_2$, $w_3$, $\alpha$, $\beta$, $\gamma$) and thresholds ($\tau$ , energy_cutoff) are hyperparameters that we plan to tune against a small held-out set of known element substitutions rather than fix by hand, so that the choice can be justified during evaluation.

# 8 Technology Stack (If Applicable)

| Category | Version | Technology / Tool |
|---|---|---|
| **Programming Language** | 3.11+ | Python |
| **Web Framework** | Latest stable | FastAPI |
| **Frontend** | Latest stable | React, TypeScript, Tailwind CSS |
| **Backend** | Latest stable | FastAPI, Celery, Redis |
| **Database** | 15+ | PostgreSQL |
| **Scientific Libraries** | Latest stable | NumPy, Pandas, SciPy, scikit-learn |
| **Deep Learning** | 2.x | PyTorch |
| **Materials Informatics** | Latest stable | pymatgen, matminer, XenonPy, Mendeleev, ASE |
| **GNN Interatomic Potentials** | Latest stable | MACE (mace-torch), M3GNet (matgl) |
| **Vector Search** | Latest stable | FAISS |
| **Explainability** | Latest stable | SHAP, Gemini API |
| **Knowledge Graph** | Latest stable | NetworkX |
| **Visualization** | Latest stable | Plotly, Matplotlib |
| **IDE / Development Environment** | – | VS Code |
| **Version Control** | – | Git, GitHub |
| **Cloud / Deployment Platform** | – | Docker, Docker Compose |
| **Operating System** | – | Ubuntu (deployment), cross-platform (development) |
| **Other Tools** | – | pytest, Alembic, mp-api, Postman |

# 9 Hardware and Software Requirements

## Hardware Requirements

| Component | Specification |
|---|---|
| **Processor** | Intel Core i5 / AMD Ryzen 5 or higher |
| **RAM** | 16 GB minimum; 32 GB recommended for larger GNN batch inference |
| **Storage** | 20 GB+ free space (cached datasets, pretrained model weights, database) |
| **GPU (If Applicable)** | NVIDIA GPU with CUDA support recommended for faster GNN scoring; CPU-only inference is possible but slower |
| **Internet Connectivity** | Required for accessing OQMD, Materials Project, and Gemini APIs |
| **Other Hardware** | None |

## Software Requirements

| Software | Version / Details |
|---|---|
| **Operating System** | Ubuntu 22.04 LTS (deployment); Windows / macOS / Linux (development) |
| **Programming Language(s)** | Python 3.11+, TypeScript |
| **Framework(s)** | FastAPI, React |
| **Database** | PostgreSQL 15+ |
| **IDE / Editor** | VS Code |
| **Libraries / Packages** | PyTorch, pymatgen, matminer, XenonPy, Mendeleev, ASE, mace-torch, matgl, FAISS, SHAP, NetworkX |
| **Other Dependencies** | Docker, Docker Compose, Redis, Node.js 18+, Git |

# 10 Design Considerations (Non-Functional)

| Design Aspect | Description |
|---|---|
| **Scalability** | GNN scoring and database validation are dispatched as asynchronous Celery tasks, so worker instances can be scaled independently of the FastAPI request layer as the candidate volume grows. |
| **Performance** | Using pretrained GNN interatomic potentials instead of running full DFT for every candidate is intended to keep per-candidate evaluation time practical enough for interactive use, following the same design choice made in the periodic-number-based crystal structure prediction work we reviewed. |
| **Security** | No personal or sensitive user data is collected. External API keys (e.g. Gemini) are stored as environment variables rather than committed to source control. |
| **Reliability** | Candidates are cross-checked against real compound databases before being surfaced, and low-confidence or energetically implausible candidates are filtered out rather than presented as firm recommendations. |
| **Maintainability** | The pipeline is split into independent modules (similarity engine, scorer, validation, explainability), so any one component – for example, swapping one GNN potential for another – can be updated without changing the rest of the system. |
| **Privacy** | The system only processes materials-science data (element and compound information); it does not collect or store personal information about users. |
| **Ethical Considerations (If Applicable)** | The system is intended as a decision-support tool, not a replacement for experimental or DFT validation. Recommendations are always shown with a confidence score and an explanation, and the interface is designed to avoid presenting low-confidence suggestions as certainties. |

# 11 Expected Outcome

By the end of the project, we expect to deliver three things: (1) an explainable AI-based element recommendation system that identifies chemically similar elements and suggests substitutes for functional materials, built around the fused similarity approach described in this report; (2) a validation module that checks generated recommendations against publicly available materials databases and literature, and reports a confidence score along with a natural-language justification for each suggestion; and (3) a working prototype and accompanying report demonstrating the framework on a small set of functional-material case studies, evaluated against the individual similarity methods (periodic-number based, co-occurrence based, property based) it combines.

We see the explainability component – generating a readable reason for each recommendation rather than only a numeric similarity or energy score – as the main point of difference between this system and the existing periodic-number-based and graph-neural-network-based crystal structure prediction methods surveyed in our literature review.

# 12 References (Optional)

1. C. Oran, R. Caputo, P. Villars, and A. Tekin, “PNcsp+: A Periodic Number-Based Crystal Structure Prediction Method Enhanced by Machine Learning,” *J. Chem. Theory Comput.*, vol. 22, pp. 3761–3771, 2026.

2. C. Oran, R. Caputo, P. Villars, H. B. Ö zcü , F. H. Canbaz, and A. Tekin, “Phase Prediction via Crystal Structure Similarity in the Periodic Number Representation,” *Inorg. Chem.*, vol. 63, pp. 20521–20530, 2024.

3. M. Kusaba, C. Liu, Y. Koyama, K. Terakura, and R. Yoshida, “Recreation of the Periodic Table with an Unsupervised Machine Learning Algorithm,” *Sci. Rep.*, vol. 11, p. 4780, 2021.

4. Q. Zhou, P. Tang, S. Liu, J. Pan, Q. Yan, and S.-C. Zhang, “Atom2Vec: Learning Atoms for Materials Discovery,” *Proc. Natl. Acad. Sci. USA*, vol. 115, pp. 6411–6417, 2018.

5. T.-C. Nguyen, V.-Q. Nguyen, V.-L. Ngo, Q.-K. Than, and T.-L. Pham, “Learning Hidden Chemistry with Deep Neural Networks,” *arXiv preprint arXiv:2101.11902*, 2021.

6. A. Y.-T. Wang, S. K. Kauwe, R. J. Murdock, and T. D. Sparks, "Compositionally Restricted Attention-Based Network for Materials Property Predictions," *npj Comput. Mater.*, vol. 7, p. 77, 2021.

7. R. Ying, D. Bourgeois, J. You, M. Zitnik, and J. Leskovec, "GNNExplainer: Generating Explanations for Graph Neural Networks," in *Proc. NeurIPS*, 2019.

8. K. T. Butler, D. W. Davies, H. Cartwright, O. Isayev, and A. Walsh, "Machine Learning for Molecular and Materials Science," *Nature*, vol. 559, pp. 547–555, 2018.

9. V. Tshitoyan, J. Dagdelen, L. Weston, A. Dunn, Z. Rong, O. Kononova, K. A. Persson, G. Ceder, and A. Jain, "Unsupervised Word Embeddings Capture Latent Knowledge from Materials Science Literature," *Nature*, vol. 571, pp. 95–98, 2019.

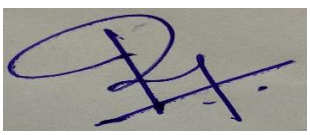
## Faculty Remarks

**Guide Remarks**

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**Project Guide**

Name: R E M Y A V S

Signature: 

**Project Coordinator**

Name: _____

Signature: ___